

Comparative Analysis of Computer Vision Algorithms for the Real-Time Detection of Digital Dermatitis
in Dairy Cows

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Abstract

Digital dermatitis (DD) is a bovine claw disease responsible for ulcerative lesions on the planar aspect of the hoof. DD is associated with massive herd outbreaks of lameness and influences cattle welfare and production. Early detection of DD can lead to prompt treatment and decrease lameness. Computer vision (CV) provides a unique opportunity to improve early detection. The study aims to train and compare applications for the real-time detection of DD in dairy cows. Eight CV models were trained for detection and scoring, compared using performance metrics and inference time, and the best model was automated for real-time detection using images and video.

Images were collected from commercial dairy farms while facing the interdigital space on the plantar surface of the foot. Images were scored for M-stages of DD by a trained investigator using the M-stage DD classification system with distinct labels for hyperkeratosis (H) and proliferations (P). Two sets of images were compiled: the first dataset (Dataset 1) containing 1,177 M0/M4H and 1,050 M2/M2P images and the second dataset (Dataset 2) containing 240 M0, 17 M2, 51 M2P, 114 M4H, and 108 M4P images. Models were trained to detect and score DD lesions and compared for precision, recall, and mean average precision (mAP) in addition to inference time in frame per second (FPS).

Seven of the nine CV models performed well compared to the ground truth of labeled images using Dataset 1. The six models, Faster R-CNN, Cascade R-CNN, YOLOv3, Tiny YOLOv3, YOLOv4, Tiny YOLOv4, and YOLOv5s achieved an mAP between 0.964 to 0.998, whereas the other two models, SSD and SSD Lite, yielded an mAP of 0.371 and 0.387 respectively. Overall, YOLOv4, Tiny YOLOv4, and YOLOv5s outperformed all other models with almost perfect precision, perfect recall, and a higher mAP. Tiny YOLOv4 outperformed all other models with respect to inference time at 333 FPS, followed by YOLOv5s at 133 FPS and YOLOv4 at 65 FPS. YOLOv4 and Tiny YOLOv4 performed better than YOLOv5s compared to the ground truth using Dataset 2. YOLOv4 and Tiny YOLOv4 yielded a similar mAP of 0.896 and 0.895, respectively. However, Tiny YOLOv4 achieved both higher precision and recall compared to YOLOv4.

26 Finally, Tiny YOLOv4 was able to detect DD lesions on a commercial dairy farm with high performance
27 and speed. The proposed CV tool can be used for early detection and prompt treatment of DD in dairy
28 cows. This result is a step towards applying CV algorithms to veterinary medicine and implementing real-
29 time DD detection on dairy farms.

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Introduction

Digital dermatitis (DD) is the most prevalent bovine infectious claw disease worldwide (Refaai et al., 2013; Logue et al., 2005; Teixeira et al., 2010). The disease is responsible for painful circumscribed ulcerative lesions on the planar aspect of the hoof around the coronary band, in the interdigital cleft, under the dew claws, and on the pastern skin (Cheli and Mortellaro, 1974; Döpfer et al., 1997). Massive herd outbreaks of lameness are caused by DD, influencing cattle welfare and production (Cramer et al., 2009; Krull et al., 2016; Jacobs et al., 2019). In addition, the disease results in major losses for the cattle industry due to decreased milk production, decreased fertility rate, and increased premature culling. (Vanhoudt et al., 2019). Early detection can lead to prompt treatment and decrease lameness within the herd (Schulz et al., 2016). Lesions are detected through visual inspection and scored using the M-stage classification system by trained investigators (Döpfer et al., 1997; Berry et al., 2012). Early identification and prompt intervention requires extensive employee training in an industry with a high employee turnover rate (Döpfer et al., 2012; Shearer and Hernandez, 2000).

Recent advances in computational power and state-of-the-art algorithms have made object detection and classification possible by applying machine learning techniques (Vaidya and Paunwala, 2019; Wu et al., 2020). Such methods for animal detection have been used to assess the health and well-being of livestock in cattle and small ruminants (Alsaad et al., 2014; Byrne et al., 2017; Gomes et al., 2016; Hovinen et al., 2008; Metzner et al., 2015; Scoley et al., 2018; Yang et al., 2018; Zaninelli et al., 2018). However, applications for the classification of health events are still rare in veterinary medicine and food production (Vinicki et al., 2018). There are many unexplored applications in veterinary medicine and an untapped potential for machine learning algorithms to predict relevant biological outcomes (Liakos et al., 2018). Thus, machine learning provides a unique opportunity to improve early detection of DD and optimized treatment plans for cattle.

Computer vision (CV) can be used to perform object detection and calculate the associated class probabilities from a series of images or videos (Szeliski, 2022). Object detection locates the presence of

an object with a bounding box, class label, and class probability (Zhao et al., 2019). The input is an image with one or more objects and the output is an image or video with zero or more class detections. The bounding box is defined by x- and y- coordinates of the center, width, height, and a class label for each object. The conditional class probability is the probability that the detected object belongs to a particular class (Zhao et al., 2019; Wu et al., 2020; Sharma and Mir, 2020).

Object detectors are mainly divided into two-stage object detectors and one-stage object detectors. The first stage of two-stage networks identifies region proposals, or subsets of the image that might contain an object. The second stage classifies the objects within the region proposals (Wu et al., 2020; Sharma and Mir, 2020). One-stage networks produce predictions for regions across the entire image using anchor boxes, and the predictions are decoded to generate the final bounding boxes for the objects. The approach skips the region proposal stage and runs detection directly over a dense sampling of possible locations (Wu et al., 2020; Sharma and Mir, 2020). Single-stage networks can be much faster than two-stage networks, but they may not reach the same level of accuracy, especially for images containing small objects.

One-stage object detectors such as You Only Look Once (YOLO) and Single Shot MultiBox Detector (SSD) in addition to two-stage object detectors such as Faster and Cascade region-based convolutional neural networks (R-CNNs) are commonly used, because of higher speed and higher accuracy compared to other computational approaches (Soviany and Ionescu, 2018). Such models are used for real-time detection of health outcomes to generate bounding boxes and class probabilities from labeled objects in images (Ünver and Ayan, 2019; Cao et al., 2017; Al-antari et al., 2020). The current study compares and contrasts five versions of YOLO: YOLOv3, Tiny YOLOv3, YOLOv4, Tiny YOLOv4, and YOLOv5s for differences in prediction performance in addition to the SSD, SSD Lite, Faster R-CNN, and Cascade R-CNN (Redmon and Farhadi, 2018; Adarsh et al., 2020; Bochkovskiy et al., 2020; Liu et al., 2016; Ren et al., 2016; Cai and Vasconcelos, 2017; ultralytics, 2022).

YOLO models have several advantages over other object detectors. First, YOLO can recognize multiple objects in a single frame (Redmon et al., 2016). It is extremely fast, because it uses a single network evaluation to simultaneously predict multiple bounding boxes and class probabilities, where other systems require thousands of network evaluations for a single image (Redmon et al., 2016). YOLO looks at the entire image during training and test time to implicitly encode contextual information about classes including their appearance and generalize representations of the objects (Redmon et al., 2016).

YOLO has achieved high performance on a high-end graphics processing unit (GPU), but does not perform fast enough on portable devices, because of the high memory requirements (Redmon, 2018; Redmon and Farhadi, 2018). Consequently, lightweight network architectures are used for constrained environments such as portable devices resulting in reduced model size and computation cost (Redmon, 2018). Tiny YOLO is the compressed version of YOLO with a simpler network structure and a reduced number of corresponding parameters (Adarsh et al., 2020). For real-time object detection, Tiny YOLO is the better option compared to YOLO, where speed is prioritized over precision or accuracy (Adarsh et al., 2020).

The YOLO models were compared on the Microsoft Common Objects in Context (MS COCO) dataset commonly used for benchmarking object detection tasks (Lin et al., 2014). The large-scale dataset contains 330K images where 200K images are annotated using 80 object categories of common objects such as a person, bicycle, and various animals, as well as more specific categories such as an umbrella, backpack, and various sports equipment (Lin et al., 2014). Additionally, MS COCO contains more small objects and more densely located objects (Lin et al., 2014). The speed of Tiny YOLOv4 is approximately eight times faster than YOLOv4 (Alexey, 2021; Bochkovskiy et al., 2020; Techzizou, 2021).

Furthermore, the speed of YOLOv5s is approximately two-and-a-half times faster than YOLOv4 (Kin-Yiu, 2020). However, the accuracy for YOLOv4 is 33% better than Tiny YOLOv4 tested on the MS COCO dataset (Alexey, 2021; Bochkovskiy et al., 2020; Techzizou, 2021). Moreover, the accuracy for

YOLOv4 is better than 11% better than YOLOv5s on the MS COCO dataset using the same image and batch size (Kin-Yiu, 2020).

A YOLOv2 model was previously trained on a database of more than 3,500 DD lesion images to detect M-stages of DD (Cerneck et al., 2020). The resulting predicting accuracy was 71% and the agreement between the human investigator and CV model was quantified as “moderate” by Cohen’s kappa during internal validation (Cerneck et al., 2020). In the external validation, the YOLOv2 model detected DD with an accuracy of 88% and agreement between a human investigator and CV model was quantified as “fair” by Cohen’s kappa (Cerneck et al., 2020).

The YOLOv3 model builds on the YOLOv2 framework by making detections at three different scales and predicts ten times more bounding boxes to improve performance on smaller objects (Redmon and Farhadi, 2018). YOLOv4 improves on the average precision (AP) by 10% and frames per second (FPS) by 12% compared to YOLOv3 on the COCO dataset (Bochkovskiy et al., 2020). For the purpose of our use case, different state-of-the-art CV models are trained using the same datasets to determine the best performing approach for fast and accurate object detection. Model performance is characterized by common benchmarking measures for speed and accuracy.

The study aims to design, develop, and implement an application for the real-time detection of DD in dairy cattle with the intention of minimizing the adverse effects of DD and lameness in all cattle by means of early detection and prompt treatments. Nine state-of-the-art CV models are trained for detection and scoring of DD, compared for performance, and implemented for real-time detection. Such tools have not yet been implemented and will help dairy farmers and producers to improve DD prevention strategies for early intervention as well as helping increase cattle welfare and production. We hope the proposed tool can be employed in combination with current practice for prevention of DD in dairy cattle. Automated detection of DD on streaming video can be used to generate treatment lists in real-time for dairy and beef cattle. Ultimately, the workflow can be deployed as a web-based application for the real-time detection of DD lesions. Such approaches can help identify high risk cattle for DD. Monitoring herds with endemic

DD for changes in lesion prevalence or severity and classifying cattle based on lesion stages can improve on-farm decision-making processes (Plummer and Krull, 2017).

Methods

Data Collection and Image Labeling

We used two different datasets for the study: the first dataset (Dataset 1) of images was initially compiled for the YOLOv2 model to detect M-stages of DD and used to compare the performance of all eight CV models (Cernek et al., 2020); and a second dataset (Dataset 2) of images was specifically compiled for the study and used to evaluate the performance of the best models for real-world application. Dataset 1 contained a collection of 2,227 images of cattle feet from midwestern United States commercial dairy farms and images were labeled for M-stages of DD (Cernek et al., 2020). Images were generated using GoPro Hero 5 Black cameras to collect MP4 video recordings of the plantar aspect of standing feet of dairy cows. Additionally, traditional and cell phone cameras were used to obtain JPG images from other studies in Wisconsin; Manitoba, Canada; and Utrecht, The Netherlands. All images were collected between June 2018 and March 2019. A rich, diverse library of images was compiled from various settings and scenarios including images of feet lifted in stand-up restraining chutes and cattle standing in the cattle housing area, in automated milking systems, or in rotary milking parlors. Dataset 2 contained a collection of 409 images of cattle feet from milking parlors in Wisconsin and images were labeled for M-stages of DD. As previously described, GoPro Hero 5 Black cameras were used to take MP4 video recordings of the backside of the hind feet at claw level. Images were taken during May 2021 of hooves lifted in the cattle housing area, automated milking system, or exiting a disinfecting foot bath.

Images were scored for M-stages of DD by a trained investigator using the M-stage DD classification system with signs of chronicity. The classification system describes various clinical stages over the course of the disease based on morphological observations between healthy (M0), active (M2), and chronic (M4)

lesions with hyperkeratosis (H) or proliferation (P) (Döpfer et al., 1997; Berry et al., 2012; Cernek et al., 2020; Anklam et al., 2022a, 2022b). For Dataset 1, the images were combined and reclassified into healthy or chronic M0/M4H lesions versus active ulcerative M2/M2P lesions while M4P lesion were removed for simplicity (Cernek et al., 2020). For Dataset 2, the images were classified into five different stages of DD: M0, M2, M2P, M4H, and M4P.

Overall, the first library (Dataset 1) of a single foot per image and two class labels included 1,177 M0/M4H class labels (936 JPG images from MP4 video recordings and 241 JPG images from camera images) and 1,050 M2/M2P class labels (660 screenshot JPG images from MP4 video recordings and 390 JPG images from camera images) and was identical to the dataset used to train the YOLOv2 model. The second library (Dataset 2) of multiple feet per image and five labels consisted of 240 M0, 17 M2, 51 M2P, 114 M4H, and 108 M4P class labels (409 screenshot JPG images from MP4 video recordings). The associated annotations for the bounding boxes and class labels were generated in Labelling version 1.8.1 using Python (Lin, 2018). Labels were converted from Pascal VOC format for R-CNN and SSD models to YOLO format for YOLO models (Tashiev, 2022).

Model Training and Performance Evaluation

The datasets were split with 90% of the images placed into a training set and 10% of the images placed into a testing set at random. For Dataset 1, the testing set contained 222 images including 117 M0/M4H and 105 M2 class labels. For Dataset 2, the testing set contained 40 images consisting of 20 M0, 4 M2, 4 M2P, 14 M4H, and 10 M4P class labels. All models were trained and tested using Google Colab and a 12GB NVIDIA Tesla K80 GPU with a batch size of 64 and subdivision size of 16 (“Google Colaboratory,” 2022). The training weights were initialized using convolutional weights that were pre-trained on ImageNet (“ImageNet,” 2020; Krizhevsky et al., 2012). The eight different computer vision models were trained until the model converged or the maximum number of batches was reached.

176 The R-CNNs were performed using ResNet-50 as the backbone feature extractor and input images were
 177 resized to 640 by 640 pixels. Faster R-CNN improved on Fast R-CNN by utilizing a Region Proposal
 178 Network to generate high-quality region proposals used by Fast R-CNN for detection (Ren et al., 2016).
 179 Cascade R-CNN extended Faster R-CNN where a sequence of detectors was trained stage by stage to
 180 leverage the output of a previous detector to train the next detector (Cai and Vasconcelos, 2017).

181 SSD uses MobileNet V1 as the backbone feature extractor and SSDLite network uses MobileNet V2. The
 182 SSD network accepts images that are resized to an input resolution of 300 by 300 pixels (Liu et al., 2016).
 183 SSDLite is similar to SSD, but implements depthwise-separable convolutions rather than regular
 184 convolution layers, which increases the speed compared to the regular SSD making it perfectly suited for
 185 use on mobile devices (Fan et al., 2018).

186 The YOLOv3 network uses a variant of the custom deep architecture Darknet, originally a 53-layer
 187 network trained on ImageNet with a 53-layer stack of neurons added for the task of detection to result in a
 188 total of a 106-layer fully convolutional underlying architecture (Redmon et al., 2016; Redmon and
 189 Farhadi, 2016, 2018). The YOLOv4 and YOLOv5 network uses CSPDarknet53 as the backbone, a spatial
 190 pyramid pooling additional module, PANet path-aggregation neck, and a YOLOv3 head (Bochkovskiy et
 191 al., 2020; ultralytics, 2022). The YOLOv4 network size height and width is 416 by 416 pixels with a
 192 learning rate of 0.001 for a maximum number of 4,000 batches. The YOLOv5 network size height and
 193 width is also 416 by 416 pixels with an initial learning rate of 0.001 for a maximum number of 1,500
 194 epochs. The best training weights were selected for model evaluation.

195 For speed, frames per second (FPS) were used to evaluate the time to predict the bounding boxes and
 196 class labels of the located objects in an image. For localization and identification, precision, recall,
 197 average precision (AP) and mean average precision (mAP) at intersection over union (IOU) of 0.5 were
 198 used as performance measures to compare between the predictions made by the CV models, and labels
 199 made by a trained investigator (the so-called 'ground truth'). The speed metric FPS calculates the ability
 200 to perform real-time object detection, while mAP and other performance measures calculate both the

ability to predict presence versus absence of the DD lesion or the localization of the object and the ability to predict the M-stage classification of the DD lesion or the identification of the class of the object into a single, unified metric. Lastly, the best CV model was validated for detection of each M-stage on video files from Dataset 2.

Results

Tiny YOLOv4 outperformed all other models with a speed of 333 FPS (Fig. 1). The closest model was YOLOv5s at 133 FPS, then SSD and SSD Lite at approximately 100 FPS, followed by YOLOv4 at 65 FPS. Cascade R-CNN and Faster R-CNN performed worse than the seven other models with a speed less than 10 FPS. For the purpose of real-time detection on a portable, stand-alone device, Tiny YOLOv4 was the best model for the use case, based on the size and speed of the implementation.

Seven of the nine CV models performed well compared to the ground truth using the first dataset with a single object per image and two class labels for object detection (Fig. 2 - 4). In addition, the seven models outperformed SSD and SSD Lite. The six models achieved an mAP between 0.964 to 0.998, whereas SSD and SSD Lite yielded an mAP of 0.371 and 0.387 respectively. Cascade R-CNN significantly outperformed Faster R-CNN for all three performance measures including precision, recall, and mAP. YOLOv4, Tiny YOLOv4, and YOLOv5s outperformed YOLOv3 and Tiny YOLOv3 for all three performance measures of precision, recall, and mAP. In addition, YOLOv3 had a lower precision but higher recall compared to Tiny-YOLOv3, but still had a slightly higher mAP. YOLOv4, Tiny YOLOv4, and YOLOv5s achieved similar precision and recall, however Tiny YOLOv4 and YOLOv5s achieved the highest mAP of 0.998 and 0.995, respectively. Overall, YOLOv4, Tiny YOLOv4, and YOLOv5s outperformed all other models with almost perfect precision, perfect recall, and a higher mAP.

Tiny YOLOv4 was an extremely fast and accurate model for real-time object detection (333 FPS and mAP of 0.998) and the best model for object detection on a portable, stand-alone device, followed by YOLOv5s (133 FPS and mAP of 0.995) and YOLOv4 (65 FPS and mAP of 0.974). Consequently,

YOLOv4, Tiny YOLOv4, and YOLOv5s were evaluated for real-world application using the second dataset of multiple objects per image and more class labels for object detection (Fig. 5 - 6). YOLOv4 and Tiny YOLOv4 performed better than YOLOv5s compared to the ground truth yielding a similar mAP of 0.896 and 0.895, respectively. However, Tiny YOLOv4 achieved both higher precision and recall. Tiny YOLOv4 was able to detect all five class labels well (Fig. 7 - 8). The model was able to detect M2P, M4H, and M4P lesions with a higher average precision compared to M2 lesions. Overall, Tiny YOLOv4 was able to detect all five M-stages of DD on video files.

Discussion

Seven of the nine CV models were able to detect DD lesions in dairy cattle from an image. For speed, Tiny YOLOv4 clearly outperformed all other models with the highest FPS. For detection, YOLOv4, Tiny YOLOv4, and YOLOv5 outperformed all other models with high precision, high recall, and the highest mAP. The next closest model was Cascade R-CNN followed by Faster R-CNN, YOLOv3, and Tiny YOLOv3. YOLOv4, Tiny YOLOv4, and YOLOv5 achieved similar precision and recall using the first dataset, but Tiny YOLOv4 yielded higher mAP. YOLOv4 and Tiny YOLOv4 achieved similar mAP using the Dataset 2, however Tiny YOLOv4 yielded higher precision and recall. Tiny YOLOv4 was the best CV model for the use case of real-time detection on streaming video. Additionally, the YOLOv3, YOLOv4, and YOLOv5 models had better performance than the YOLOv2 model using the same input data (Cernek et al., 2020). As a proof-of-concept, the Tiny YOLOv4 model was able to detect the different M-stages of DD on video files.

For future application, the comparison of CV models for a given task can help identify which approaches perform best for a given situation (Cao et al., 2019; Jia et al., 2020; Liu et al., 2020; Xu et al., 2020). For instance, the R-CNN models, including Faster, Mask, and Cascade R-CNN, provide better detection performance when hooves were in close proximity, such as in densely populated dairy barns or feedlots. For applications where hooves were clearly separated from each other, such as a milking parlor, the YOLO models were a superior approach for the detection of DD lesions.

For Dataset 2, Tiny YOLOv4 was able to detect all five class labels well. However, the model was able to detect M2P, M4H, and M4P lesions better than M2 lesions with a higher average precision. This was probably caused by unbalanced classes of data in the second dataset with a relatively small number of M2 in the training set. Unbalanced datasets where the number of instances for class labels, such as M2, is less than the other class labels can create problems with regards to overall prediction performance of the CV model (Oksuz et al., 2021). Adding images to the dataset for infrequent classes will improve the model performance. The difference in distance between the camera and the object can create inaccurate object detection performance when using YOLO models (Redmon and Farhadi, 2018).

The CV models applied during this project were able to identify and classify DD lesions on commercial dairy farms with high accuracy, in real-time, and at high speed. This result is a step in applying CV algorithms to veterinary medicine and implementing real-time DD detection on dairy farms. The proposed CV tool can be used for early detection and prompt treatment of DD in dairy cattle and has been extended to beef cattle (unpublished data). The trained model may be applied to different cattle breeds and various locations. The CV models for DD detection can be expanded to include more images from more farms for implementation on different farms. The YOLO framework can classify multiple claw diseases simultaneously in the future after training the models using more images of claw lesions. For increased generalization, the model requires additional images originating from additional cattle breeds, animal species, and geographical regions for robust predictions. CV has the potential to be used in precision livestock farming and high-throughput phenotyping applications. Such applications include animal identification and tracking, behavior monitoring, and phenotype classification and recognition (Fernandes et al., 2020). Applications of CV in animal production systems and veterinary sciences are a growing research area (Fernandes et al., 2020; Li et al., 2021; Wurtz et al., 2019).

The M-stage classification system in clinical medicine is beneficial for treatment decision-making and monitoring treatment success where advanced lesions are associated with lameness and increased transmission (Plummer and Krull, 2017). Inspection of cleaned feet with a swiveling mirror and a

powerful headlamp in the milking parlor is useful for both farmers and researchers may improve the discrimination of each M-stage, especially for early lesions (Relun et al., 2011). However, M1 lesions are infrequent and are commonly misclassified by scorers as M0 or M4 lesions because M1 lesions are too small to detect (Relun et al., 2011; Solano et al., 2017). Furthermore, these skin defects come and go generally within 10-day intervals (Kofler et al., 2019). Topical antibiotics are used to treat only active M2 lesions, accelerating the transition to the chronic stage (Cramer and Solano, 2023). Therefore, identifying the M2 stage of the disease compared to M1 stage was sufficient in this study to accurately detect active lesions and inform the necessary treatment and control strategy.

The TensorFlow 1.x framework used in the study can be expanded to other open-source machine learning frameworks for deployment and overall ease of use. TensorFlow 2.x offers better usability and high performance (NVIDIA Corporation, 2022; tensorflow, 2017). TensorFlow models can be converted to TensorFlow Lite models reducing model size and power consumption and subsequently, they can be deployed on edge devices like mobiles or microcontrollers (tensorflow, 2021). PyTorch framework is quite fast and extremely efficient compared to some of the alternatives (Paszke et al., 2019; pytorch, 2021). YOLOv3 has been implemented in PyTorch by Ultralytics. YOLOv5 is the PyTorch implementation of YOLOv4 and achieves similar speed and accuracy, but reduced training time (ultralytics, 2022, 2021). The PyTorch implementation of YOLOv3 or different size models of YOLOv5 can be evaluated to see if there is an improvement in performance or speed. Lastly, YOLOv4 and YOLOv5 can be updated to the most recent version of each implementation, namely YOLOv7 and YOLOv8, providing an increase in accuracy and efficiency (Jocher, 2023; Jocher et al., 2023; Wang et al., 2022; Wong, 2023). The selection of object detection algorithms in this study was based on the available documentation and code samples for model embedding on an edge device, ultimately choosing the Tiny YOLOv4 model because of its superior performance in real-time object detection tasks (Dat, 2023; Luxonis, 2023a, 2023b, 2023c, 2023d, 2023e, 2023f, 2023g, 2022a, 2022b).

Precision and recall were selected over other metrics including sensitivity and specificity to evaluate the performance of machine learning models. Precision and recall and its composite performance measures such as AP and mAP provide numerous advantages in the context of object detection, while sensitivity and specificity are widely used in binary classification tasks. In object detection, the focus is correctly identifying objects of interest (true positives) while minimizing false detections (false positives).

Precision and recall precisely quantifying the accuracy of positive predictions and the coverage of true positives, respectively, which are particularly important in object detection tasks where correctly identifying and localizing objects is the primary objective. Sensitivity and specificity, on the other hand, also consider true negative predictions, which may be less relevant in this context, since there exists infinitely many bounding boxes that should not be detected for a given image (Padilla et al., 2020).

Accessibility to CV tools can promote the growth of such techniques in veterinary medicine and reduce adoption thresholds for the models. The CV models can be extended to portable platforms, including iOS or Android application for mobile use, as well as using a docker container for deployment of CV models as cloud-based applications (Docker, 2021). The implementation of such CV models in hand-held devices and in collaboration with cattle professionals will generate a rich, diverse library of images for the optimization and validation of CV models. Ultimately, the proposed CV tool will improve animal welfare and increase production for large-scale cattle facilities as an example for other applications in veterinary medicine and agriculture.

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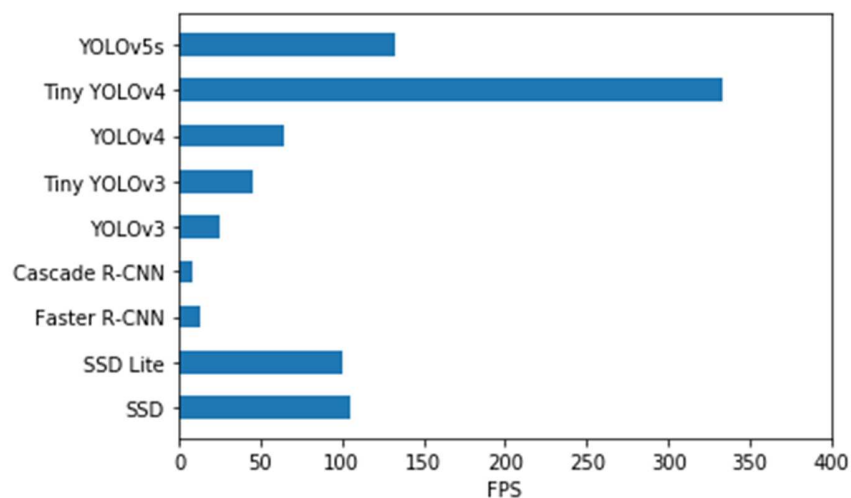
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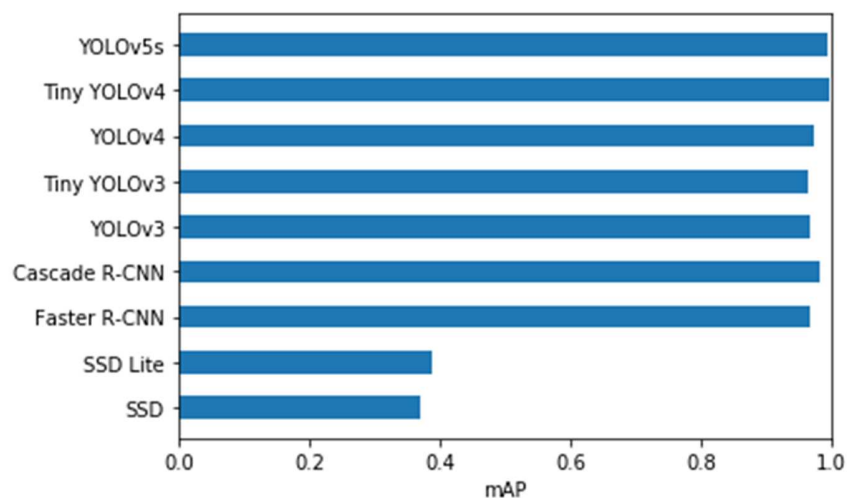


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548 Fig. 1. Inference time of the nine CV algorithms for the testing set in Dataset 1 (n= 2,227). Inference time
549 was measured in frames per second (FPS) and the nine CV models include the five latest versions of You
550 Only Look Once (YOLO) models: YOLOv3, Tiny YOLOv3, YOLOv4, Tiny YOLOv4, and YOLOv5s in
551 addition to Single-Shot Multibox Detector (SSD), SSD Lite, Faster Region-Based Convolutional Neural
552 Networks (R-CNN), and Cascade R-CNN.

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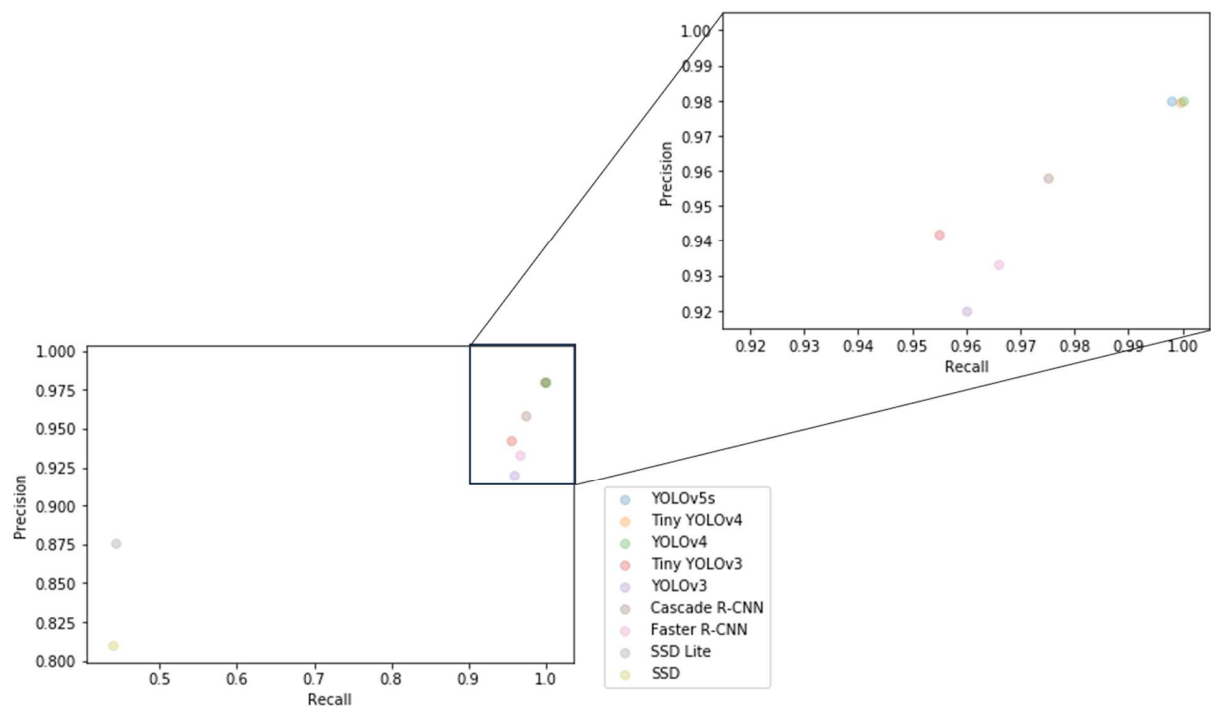


555

556 Fig. 2. Performance of the nine CV algorithms for the testing set in Dataset 1 (n= 2,227). Performance
557 was measured in mean average precision (mAP) and the nine CV models include the five of the latest
558 versions of You Only Look Once (YOLO) models: YOLOv3, Tiny YOLOv3, YOLOv4, Tiny YOLOv4,
559 and YOLOv5s in addition to Single-Shot Multibox Detector (SSD), SSD Lite, Faster Region-Based
560 Convolutional Neural Networks (R-CNN), and Cascade R-CNN.

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564 Fig. 3. Precision and recall of the nine CV algorithms (bottom-left) with inset of the top seven CV model
565 (top-right) for the testing set in Dataset 1 (n= 2,227). Performance was measured in precision and recall
566 and the nine CV models include the five of the latest versions of You Only Look Once (YOLO) models:
567 YOLOv3, Tiny YOLOv3, YOLOv4, Tiny YOLOv4, and YOLOv5s in addition to Single-Shot Multibox
568 Detector (SSD), SSD Lite, Faster Region-Based Convolutional Neural Networks (R-CNN), and Cascade
569 R-CNN.

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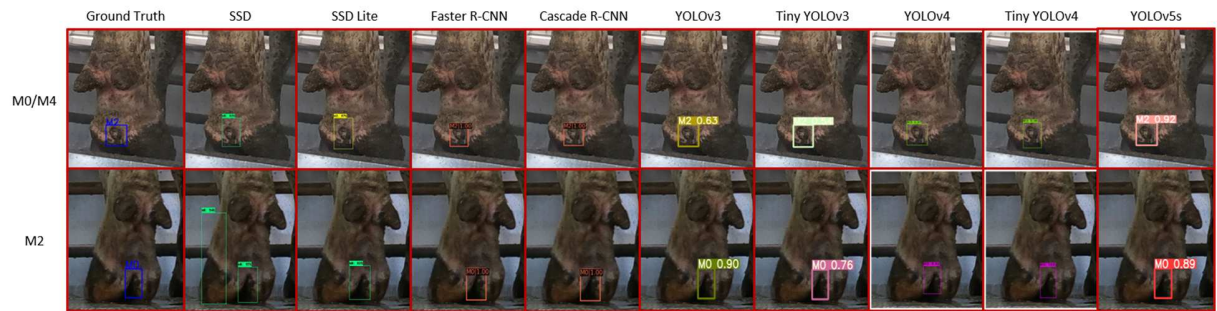
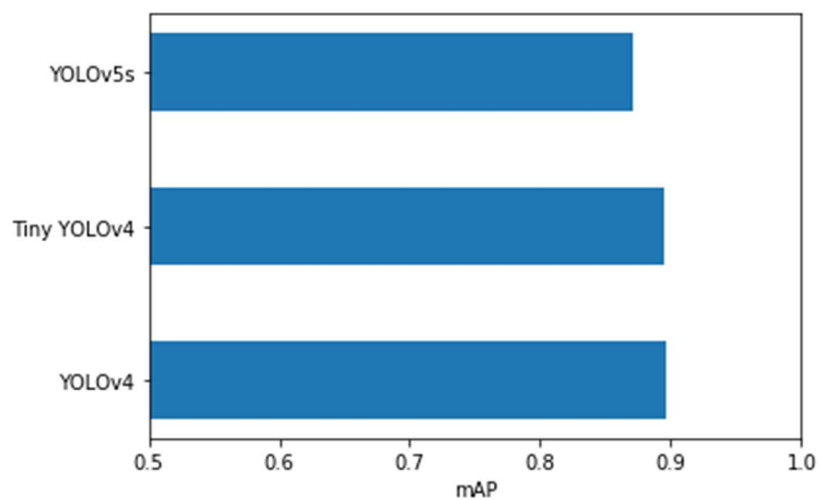


Fig. 4. Bounding box predictions of M0/M4 and M2 by the nine CV algorithms for the testing set in Dataset 1 (n= 2,227). The nine CV models include the five latest versions of You Only Look Once (YOLO) models: YOLOv3, Tiny YOLOv3, YOLOv4, Tiny YOLOv4, YOLOv5s in addition to Single-Shot Multibox Detector (SSD), SSD Lite, Faster Region-Based Convolutional Neural Networks (R-CNN), and Cascade R-CNN compared to labels by a trained investigator ('Ground Truth').

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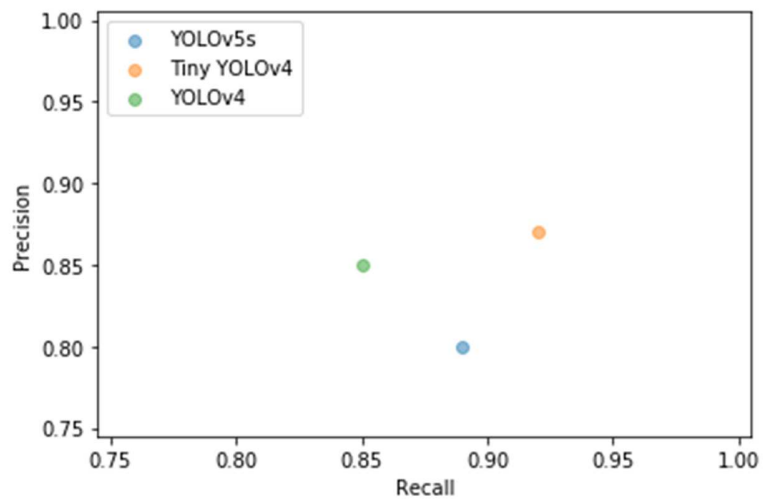


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581 Fig. 5. Performance of the top three CV algorithms for the testing set of Dataset 2 (n= 409). Performance
582 was measured in mean average precision (mAP) and the CV models include the three of the latest
583 versions of You Only Look Once (YOLO) models: YOLOv4, Tiny YOLOv4, and YOLOv5s.

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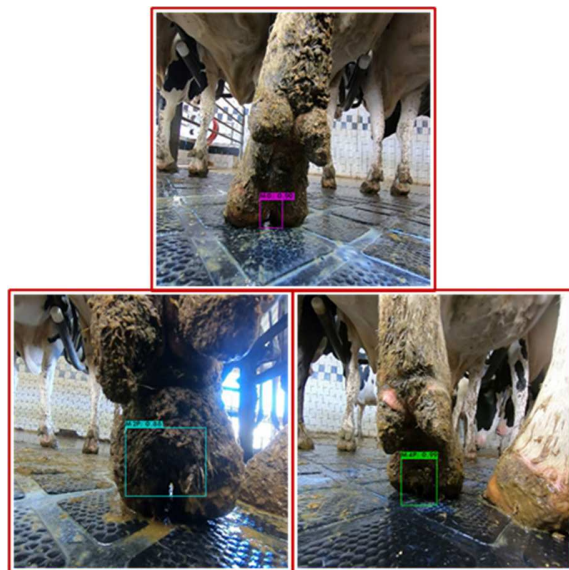
587 Fig. 6. Precision and recall of the top three CV algorithms for the testing set of Dataset 2 (n= 409).

588 Performance was measured in precision and recall and the CV models include the three of the latest

589 versions of You Only Look Once (YOLO) models: YOLOv4, Tiny YOLOv4, and YOLOv5s.

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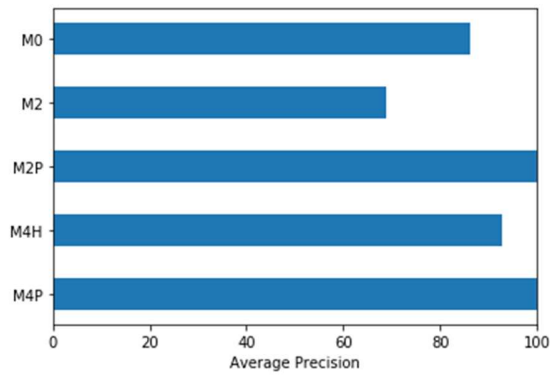
593 Fig. 7. Bounding box predictions of M-stages by Tiny YOLOv4 for the testing set in Dataset 2.

594 Predictions were made for each of the M-stages: M0 (top-center; magenta), M2, M2P (bottom-left; cyan),

595 M4H, and M4P (bottom-right; green).

596

597



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599 Fig. 8. Average precision of the five M-stages by Tiny YOLOv4 for the testing set in Dataset 2 (n= 409).

600 Performance was measured in average precision and calculated for each of the five M-stages: M0, M2,

601 M2P, M4H, and M4P.

602